

Lecture 3: Elasticities & Taxes — the Supply & Demand model II (Ch. 2 recap · §6.4)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, §6.4 Elasticities (own-price, cross-price, income; supply elasticity) — building on the **Chapter 2** supply-and-demand model (see Lecture 2). *The lecture deck also teaches taxes on goods & services — that part draws on the textbook's §7.5 (unit tax, tax incidence), covered here because the deck foregrounds it.*

Theme of the lecture (Butera — from the deck, titled "*Supply and Demand model*"). Chapter 2 taught us to **move the curves around**; this lecture adds **precision**. The *"shape"* of a demand or supply curve — its *elasticity* — decides how much a shock (or a tax) shows up as a change in price versus a change in quantity. The deck's three goals: (1) understand what demand & supply elasticities mean, (2) derive them from demand/supply curves, and (3) use them to analyse the **effects of taxes**.

★ **The heart of the lecture:** elasticity = % responsiveness — "by what % does quantity change when price rises 1%." It's a **unit-free** number that summarises a curve's **shape**, so it's comparable across markets and time. The **shape decides how a shock splits into price vs quantity**: with **inelastic** demand a shock/tax hits **price**; with **elastic** demand it hits **quantity**. And a **tax** drives a **wedge** between the price buyers pay and sellers receive — **who bears it depends on the relative elasticities**, and (in perfect competition) **it doesn't matter who is legally taxed**.

The red thread. For any market ask: *how responsive is quantity to price (elastic or inelastic)?* — because that single question answers **how a supply shock moves the market**, **how a firm should price**, and **who really pays a tax**. Elasticity is the "shape" number that ties all three together.

Key concepts / "modes" to use

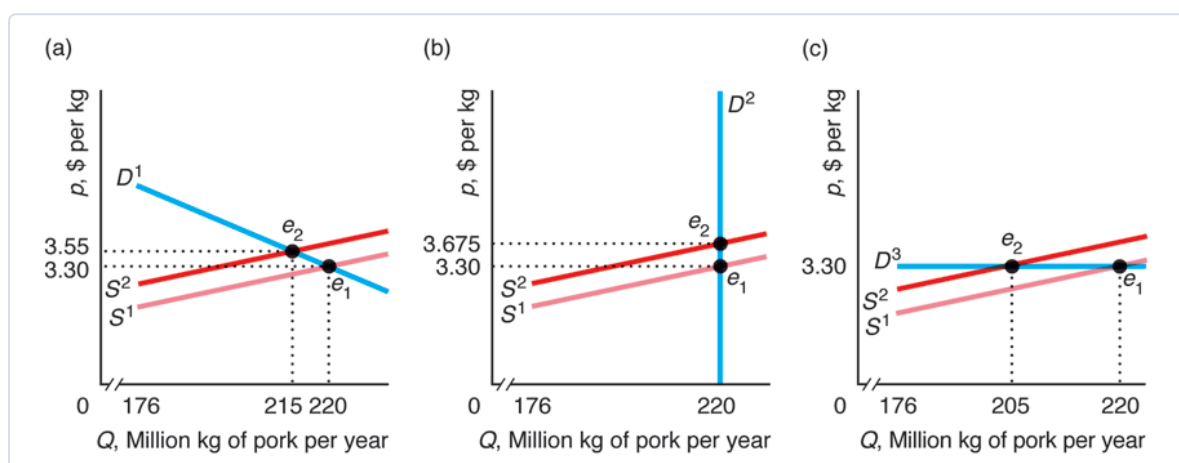
- **Elasticity** — the % change in one variable ÷ % change in another (a unit-free number).
- **Price elasticity of demand (E_D)** — % change in quantity demanded ÷ % change in own price (negative).
- **Elastic / inelastic / unit-elastic** — $|E| > 1$ / $|E| < 1$ / $|E| = 1$.
- **Point-elasticity trick** — $E_D = (dQ/dp) \cdot (p/Q)$; on a linear curve elasticity **changes along the curve** (constant-elastic curves don't).
- **Cross-price elasticity** — substitutes (+) vs complements (−).
- **Income elasticity** — normal (+) vs inferior (−); **Engel curves / Engel's law**.
- **Supply elasticity** — % change in quantity supplied ÷ % change in price (positive).
- **Tax wedge & tax incidence** — $p_b = p_s + t$; who bears the burden depends on **relative elasticities**; legal $\neq$ economic incidence; unit vs ad-valorem tax; **deadweight loss**.

Section-by-section main points

Why do we care about elasticity? (deck)

Core: the **shape** of demand/supply curves determines **how much a shift in one moves the equilibrium** — and "shape" is summarised by **elasticity**. A steep (inelastic) curve and a flat (elastic) curve respond to the *same* shock very differently.

- **The motivating questions (deck):** Do "sin taxes" (cigarettes, sugar) hurt the poor? Why do firms **raise prices** even when it looks crazy (Netflix, Amazon Prime)? To what extent are **taxes on producers "passed on"** to consumers? Every answer runs through elasticity.
- **The visual intuition:** how far a **supply shift** moves price vs quantity depends entirely on the **slope/elasticity of demand** — see the figure below (perfectly inelastic demand → only price moves; perfectly elastic demand → only quantity moves).



How to read it (why elasticity matters). All three panels apply the **same supply shift** $S^1 \rightarrow S^2$. The **only** thing that differs is the **demand curve's shape**:

- **(a) Ordinary downward-sloping demand:** the shock splits into **both** a higher price *and* a lower quantity.
- **(b) Perfectly inelastic demand (vertical):** quantity can't respond, so the shock lands **entirely on price** ($3.30 \rightarrow 3.675$). *This is the "necessity with no substitutes" case.*
- **(c) Perfectly elastic demand (horizontal):** price can't move, so the shock lands **entirely on quantity** ($220 \rightarrow 205$).
- **Take-away:** the **more inelastic** the demand, the more a shock (or tax) shows up as **price**; the **more elastic**, the more it shows up as **quantity**. *This one picture is the whole reason we measure elasticity.*

★ Price elasticity of demand — definition

Core: the **price elasticity of demand** measures how sensitive quantity demanded is to the good's own price — the **% change in quantity** ÷ **% change in price**. It is a **pure (unit-free) number** and is **negative** (demand slopes down: price up → quantity down).

PRICE ELASTICITY OF DEMAND

$$E_D = (\Delta Q / Q \times 100) \div (\Delta P / P \times 100) = \% \Delta Q \div \% \Delta P$$

Reads: how responsive quantity demanded is to a price change — "a 1% rise in price p leads to an E_D % change in quantity q ." — $\Delta Q/Q \times 100$ = the percentage change in quantity demanded · $\Delta p/p \times 100$ = the percentage change in price · E_D = the ratio (a unit-free number, negative for a normal good). The $\times 100$ s cancel, so it's just % ΔQ over % Δp .

- **Why unit-free matters (the point of the whole concept):** raw slopes aren't comparable — "1,000 tons less rice" means something different in **Vietnam** vs **Bhutan**, and depends on the currency. A percentage measure is **comparable across markets, currencies and time**.
- **Why negative:** demand curves slope **down**, so ΔQ and Δp have opposite signs → the ratio is negative. (When people say a "higher" elasticity they usually mean a bigger **absolute value** $|E_D|$.)
- **Worked example:** a café raises its coffee price **10%** and sells **8% fewer** cups → $E_D = -8\% \div +10\% = -0.8$. Since $|-0.8| < 1$, demand is **inelastic** — revenue actually *rises* (price up 10%, quantity down only 8%). (Full step-by-step computations are in the next section.)

Computing elasticity — and the point-elasticity trick

Core: plug the two percentage changes into the formula. **On a linear demand curve the elasticity is not constant** — it depends on *where* you evaluate it, via $E_D = (dQ/dp) \cdot (p/Q)$.

- **Simple example (textbook Table 6.2):** quantity falls $6 \rightarrow 4$ while price rises $1 \rightarrow 2$. Then

WORKED — SIMPLE EXAMPLE

$$E_D = ((4-6)/6) \div ((2-1)/1) = (-2/6) \div (1) = -1/3$$

Reads: a demand that is **inelastic** — $(4-6)/6 = -0.33$ = the % fall in quantity (−33%) · $(2-1)/1 = 1$ = the % rise in price (+100%) · $-1/3$ = quantity falls **only 1/3 as fast** as price rises → **inelastic** demand.

- **A policy-style example (soda, Allcott et al. 2019):** own-price elasticity of sugary soda ≈ -1.37 . If price rises 5%, quantity changes by $E_D \times 5\% = -1.37 \times 5 = -6.85\%$ (a ~7% fall). To cut demand 15%, price must rise $-15 \div -1.37 \approx 11\%$. → *this is exactly the sum a policymaker designing a soda tax needs.*
- **The point-elasticity trick (linear demand):** rewrite the formula as

POINT ELASTICITY

$$E_D = (\Delta Q/\Delta p) \cdot (p/Q) = (dQ/dp) \cdot (p/Q)$$

Reads: elasticity = **slope of the demand function** × the **price-to-quantity ratio at your chosen point** — dQ/dp = the **slope** (how many units Q changes per €1 of price; constant for a straight line) · p/Q = the **price-to-quantity ratio** where you evaluate it (this is what varies along the curve).

- **Worked ($Q = 10 - 2p$):** here $dQ/dp = -2$ (always). But the **elasticity varies**:
 - at $p = 2.5$ (so $Q = 5$): $E_D = -2 \times 2.5/5 = -1$ (**unit elastic**).
 - at $p = 4$ (so $Q = 2$): $E_D = -2 \times (p/Q) = -2 \times 4/2 = -4$ → **elastic**. (Δ The textbook misprints this as $-2 \times 2/2 = -2$ — it accidentally puts $Q=2$ in the numerator instead of $p=4$; the correct p/Q is $4/2 = 2$, so $E_D = -4$.)

- **The pattern: demand is more elastic at higher prices** — at $p=2.5$ it's unit-elastic (-1), at $p=4$ it's elastic (-4). A fixed unit change is a bigger *percentage* change when Q is small and p is large.
- **Constant-elasticity demand (contrast):** the form $Q = a \cdot p^\epsilon$ has the **same** elasticity ϵ at *every* price (used in empirical estimation via $\ln Q = \ln a + \epsilon \cdot \ln p + \text{error}$). So: **linear** $\rightarrow$ **elasticity changes along the curve**; **constant-elastic** $\rightarrow$ **it doesn't**.
- **Worked example:** with $Q = 100 \cdot p^{-1.2}$, the elasticity is -1.2 whether you evaluate at $p = \text{€}1$ ($Q = 100$), $p = \text{€}5$ ($Q \approx 15$), or $p = \text{€}50$ — it never changes. (Contrast the linear $Q = 10 - 2p$ above, where it swung from -1 to -4 .)

Elastic, inelastic, unit-elastic — and why it matters

Core: we name the ranges of elasticity, and each has a sharp economic meaning for how supply shocks split into price vs quantity.

Name	Level (of E_D)	Meaning	Curve looks...
Inelastic	$-1 < E_D < 0$ ($ E < 1$)	$\% \Delta Q$ smaller than $\% \Delta p$	relatively steep
Unit elastic	$E_D = -1$	$\% \Delta Q$ equals $\% \Delta p$	—
Elastic	$E_D < -1$ ($ E > 1$)	$\% \Delta Q$ bigger than $\% \Delta p$	relatively flat

- **Why it matters (the punchline):** with **inelastic** demand, **supply shocks translate into big price changes** (little quantity response) — critical for e.g. **cocoa** growers in Ghana/Côte d'Ivoire (world cocoa elasticity ≈ -0.19 to -0.96 ; policy point estimate -0.34). With **elastic** demand, shocks translate into **quantity** changes.
- **What makes demand elastic** — **substitutes**. The single biggest driver is **availability of close substitutes**: pharmaceuticals with few substitutes are **inelastic** (people pay whatever for a needed medicine); goods with many substitutes are **elastic**.
- **Aggregation level matters:** a *specific brand* is more **elastic** than the *category* (more substitutes for one brand). US beer: **product-level** median elasticity ≈ -4.74 , but beer **overall** ≈ -0.60 .

Cross-price elasticity — substitutes vs complements

Core: the **cross-price elasticity** measures how demand for good **A** responds to the price of a *different* good **B** — and its **sign** tells you the relationship.

CROSS-PRICE ELASTICITY

$$E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B \rightarrow > 0 = \text{substitutes} \cdot < 0 = \text{complements}$$

Reads: how the quantity of **A** responds to a price change in **B** — $\% \Delta Q_A$ = percentage change in demand for good A · $\% \Delta p_B$ = percentage change in the price of good B · **sign:** **positive** $\rightarrow$ **substitutes** (Heineken dearer $\rightarrow$ more Carlsberg bought), **negative** $\rightarrow$ **complements** (phones dearer $\rightarrow$ fewer phone cases). *The larger the positive value, the closer the substitutes.*

- **Worked example:** Danish spirits get 10% dearer → Swedish demand for spirits changes by $E_{\text{cross}} \times 10\% = +0.3 \times 10 = +3\%$ (Swedes buy **more at home** instead of driving to Denmark) → **positive** → **substitutes** (Asplund 2007). *Flip it: if smartphones get 10% dearer and phone-case demand falls 6%, $E_{\text{cross}} = -6 \div 10 = -0.6$ → negative → complements.*
- **Why it's used:** gauging **competitive pressure** and evaluating **mergers** (competition authorities ask how closely two products substitute). (*Deck aside: cross-price elasticity of cigarettes & alcohol ≈ -1 — strong complements, Krauss et al. 2014.*)

Income elasticity — normal vs inferior goods, Engel curves

Core: the **income elasticity** measures how demand responds to **income** — its **sign** distinguishes **normal** from **inferior** goods.

INCOME ELASTICITY

$$E_{\text{income}} = \% \Delta Q \div \% \Delta I \rightarrow > 0 = \text{normal good} \cdot < 0 = \text{inferior good}$$

Reads: how quantity demanded responds to an income change — $\% \Delta Q$ = percentage change in quantity · $\% \Delta I$ = percentage change in income · **sign: positive → normal** (demand rises with income), **negative → inferior** (demand falls as income rises — e.g. instant noodles, store-brand goods).

- **Worked example:** a household's **income rises 10%** and its spirits purchases rise 14% → $E_{\text{income}} = +14 \div +10 = +1.4$ → **normal and income-elastic** (a "luxury," > 1) (Asplund 2007). *Flip it: income +10% but instant-noodle demand -5% → $E_{\text{income}} = -5 \div 10 = -0.5$ → inferior.*
- **Nuance:** a good can be **normal at low incomes and inferior at higher ones** (hostels for a student: more nights as income rises, then switch to hotels). Tastes vary by person.
- **Uses:** long-term **projections** (healthcare, appliances in low-income countries) and predicting the **business cycle** (pharma is income-insensitive → "defensive stock").
- **Engel curves / Engel's law (Ernst Engel, 1800s Prussian statistician):** plot demand (vertical) against income (horizontal) — **positive slope for normal goods, negative for inferior**. **Engel's law:** the **share of income spent on food falls** as income rises → the income elasticity of food is **less than 1**.

Supply elasticity — same principle

Core: the **price elasticity of supply** measures how sensitive quantity *supplied* is to price — same formula, but **positive** (supply slopes up).

PRICE ELASTICITY OF SUPPLY

$$E_S = \% \Delta Q_S \div \% \Delta p \quad (\text{positive})$$

Reads: how responsive quantity supplied is to a price change — $\% \Delta Q_S$ = percentage change in quantity supplied · $\% \Delta p$ = percentage change in price · E_S = the ratio (**positive:** higher price → more supplied). A **steep** supply curve is **inelastic** (hard to expand output quickly); a **flat** one is **elastic**.

- **Worked example:** the market price of new flats rises **20%**, but the quantity of new flats supplied rises only **4%** this year → $E_S = +4 \div +20 = +0.2$ → **inelastic** supply (you can't build quickly). Give it several years and builders respond more → supply becomes **more elastic**.

Elasticity over time — short run vs long run

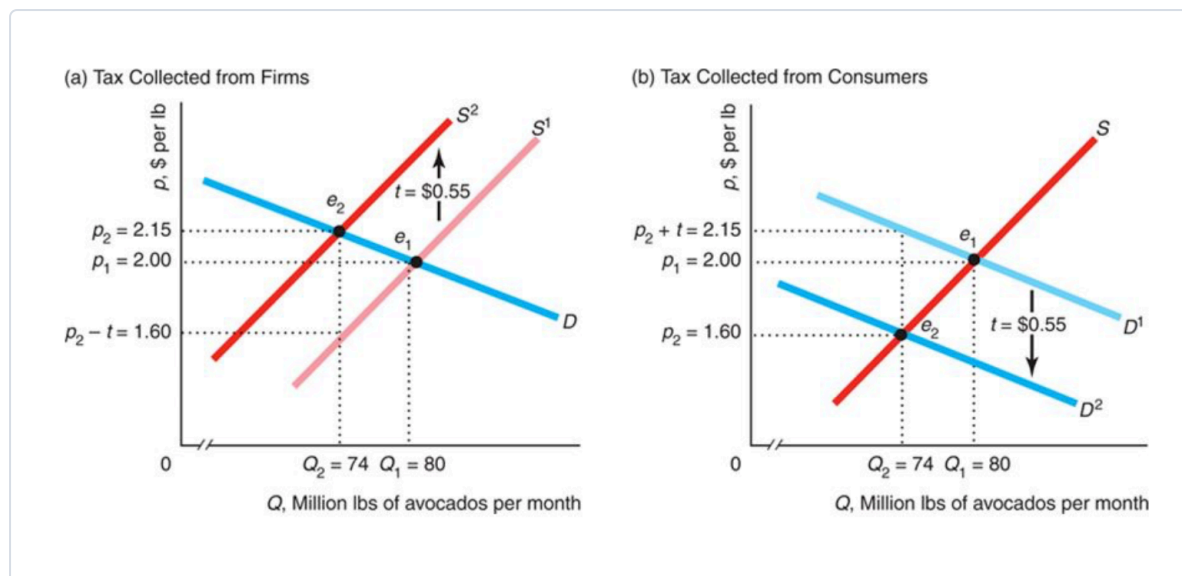
Core: demand (and supply) are usually **more elastic in the long run** than the short run, because people have **more time to adjust / find substitutes**.

- **Electricity (Labandeira et al. 2017):** ≈ -0.24 short-run vs -0.6 long-run — over time consumers buy more efficient equipment, change habits.
- **The deck's cases:** a drop in gas prices does little in the short run but more in the long run; likewise **computers**. *Exam framing: "in the short run demand is inelastic (people are locked in); given time they substitute, so long-run demand is more elastic."*

★ Taxes on goods & services — the wedge, incidence & equivalence (deck; textbook §7.5)

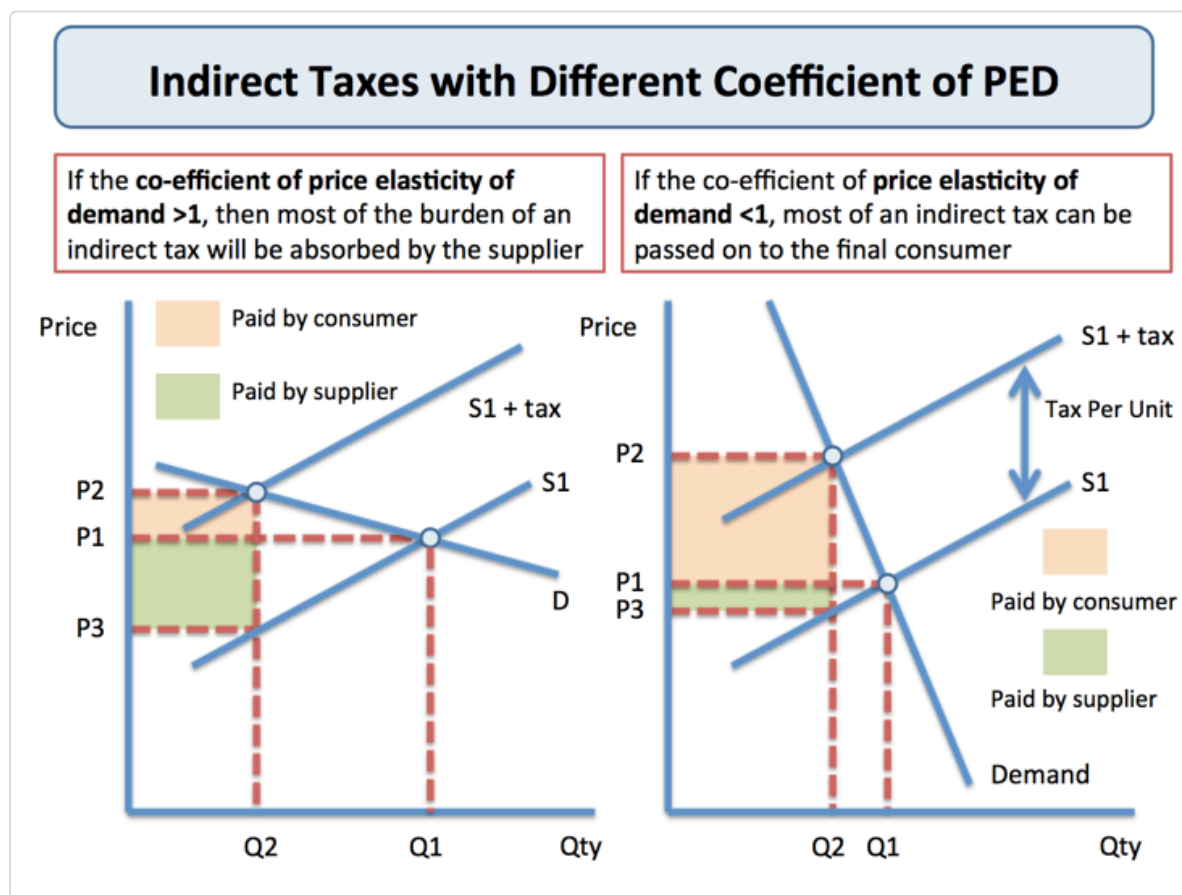
Core: a tax drives a **wedge** between the price **buyers pay** (p_b) and the price **sellers receive** (p_s): $p_b = p_s + t$. It raises the buyer price, lowers the seller price, cuts the quantity traded, and creates a **deadweight loss**. The split of the burden — **tax incidence** — depends on the **relative elasticities**, and (under perfect competition) **not** on who is legally taxed.

- **Two types of tax (deck):**
 - **Unit (specific) tax** — a fixed **amount t per unit** (per litre of petrol, per cigarette).
 - **Ad valorem tax** ("sales tax"/VAT) — a **fraction α of the price/spend** (government keeps a share of each euro spent).
- **The wedge (worked unit-tax example, $t = €3$):** with $Q_D = 23 - p_b$, $Q_S = -1 + p_s/2$, and $p_b = p_s + 3$, solving gives $p_s = 14$, $p_b = 17$, $Q = 6$. Buyers pay **€3 more** than sellers receive; quantity falls from the no-tax equilibrium.
 - **Tax revenue** $= t \times Q = 3 \times 6 = €18$. The rest of the surplus lost is **deadweight loss** (here $3 \times (7-6)/2 = €1.5$) — trades that *would* have happened but now don't.
- **Legal incidence $\neq$ economic incidence (the key result).** It **doesn't matter** whether the tax is collected from **firms** (supply shifts up by t) or **consumers** (demand shifts down by t) — the quantity, the prices actually paid/received, and the DWL are **identical**. *All that matters is that the tax drives a wedge.* (Perfect-competition assumptions make this exact: no transaction costs, perfect information.)



How to read it (tax equivalence). Both panels impose the **same \$0.55 tax** on avocados. (a) collects it from **firms** → the supply curve shifts **up** by t ; (b) collects it from **consumers** → the demand curve shifts **down** by t . In **both** cases the result is identical: **Q falls to 74**, **buyers pay 2.15**, **sellers receive 1.60**, and the wedge between them is exactly **\$0.55**. → *The economics doesn't care who writes the cheque.*

- **What does decide the split: relative elasticities.** The more **inelastic** side of the market bears **more** of the tax (it can't escape by changing quantity). The more **price-sensitive (elastic)** side is affected less on price, **more** on quantity.



How to read it (incidence & elasticity). The tax raises the supply curve to $S + \text{tax}$. The **new higher price P_2** is what **consumers** pay; **P_3** is what **suppliers** keep. The split of the rectangle between them is

the **incidence**:

- **Left — elastic demand ($PED > 1$):** consumers easily cut back, so firms **can't raise the price much** → the **supplier absorbs most** of the tax.
- **Right — inelastic demand ($PED < 1$):** consumers keep buying regardless, so the firm **passes most of the tax through** → the **consumer bears most** of it.
- **Policy take-aways:** (1) "**sin taxes**" on inelastic goods (cigarettes, alcohol) fall mostly on **consumers** and raise a lot of revenue with **little DWL** (quantity barely moves) — which is *why* governments like them, and why they can be **regressive** (hit the poor). (2) We **prefer taxes on inelastic bases** — they distort choices least.
- **Subsidies are the mirror image:** $p_s = p_b + s$ — a subsidy lowers the buyer price, raises the seller price, **expands** quantity above the efficient level, and also creates a **deadweight loss** (units produced that cost more than consumers value them). Its incidence, too, depends on elasticities (inelastic demand — e.g. emergency care, primary education — → small DWL).
 - **Worked example (subsidy $s = €3$):** same market ($Q_D = 23 - p_b$, $Q_S = -1 + p_s/2$) but now $p_s = p_b + 3$ → solving gives **$p_b = 15$, $p_s = 18$, $Q = 8$** — buyers pay **less** and sellers get **more**, and quantity **rises** above the no-subsidy level ($6 \rightarrow 8$). The subsidy costs the government $s \times Q = 3 \times 8 = €24$, part of which is **deadweight loss**.

Cases & examples

Read each as a worked use of an elasticity number (or a tax result).

- **Sugary-soda tax (Allcott et al. 2019).** Own-price elasticity ≈ -1.37 (elastic). A 5% price rise → $\sim 7\%$ fall in quantity; to cut demand 15% you'd raise price $\sim 11\%$. → *Illustrates: elasticity as the **design tool for a corrective ("sin") tax**.*
- **World cocoa demand.** Inelastic (≈ -0.19 to -0.96 ; policy estimate -0.34). → *Illustrates: **inelastic demand** → **supply shocks hit price**, so bad harvests swing incomes for growers in Ghana / Côte d'Ivoire.*
- **US beer — brand vs category (Miller et al. 2021).** Product-level elasticity ≈ -4.74 , category ≈ -0.60 . → *Illustrates: **aggregation** — a single brand has close substitutes (elastic), the category doesn't (inelastic).*
- **Sweden–Denmark cross-border alcohol (Asplund et al. 2007).** Swedes drive to Denmark for cheaper drink. Estimated for spirits: **own-price** -1.3 (elastic), **cross-price (Danish price)** $+0.3$ (substitute), **income** $+1.4$ (normal good). → *Illustrates: **all three elasticities at once** — own, cross, income — from real data.*
- **Electricity demand (Labandeira et al. 2017).** ≈ -0.24 short-run vs -0.6 long-run. → *Illustrates: demand is **more elastic in the long run** (time to adjust equipment/habits).*

Formula sheet — quick reference

Quantity	Formula	Sign / meaning
Price elasticity of demand	$E_D = \% \Delta Q_D \div \% \Delta p$	negative ; $ E > 1$ elastic, $ E < 1$ inelastic
Point elasticity (linear)	$E_D = (dQ/dp) \cdot (p/Q)$	varies along the curve; more elastic at higher p
Constant-elastic demand	$Q = a \cdot p^\epsilon \rightarrow \ln Q = \ln a + \epsilon \ln p$	elasticity = ϵ everywhere
Cross-price elasticity	$E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B$	+ substitutes, – complements
Income elasticity	$E_{\text{income}} = \% \Delta Q \div \% \Delta I$	+ normal, – inferior ; food < 1 (Engel)
Supply elasticity	$E_S = \% \Delta Q_S \div \% \Delta p$	positive
Tax wedge	$p_b = p_s + t$	buyers pay more, sellers get less; DWL
Tax revenue	$t \times Q_{\text{taxed}}$	rectangle; rest of loss = deadweight loss
Subsidy wedge	$p_s = p_b + s$	expands Q above efficient; also a DWL

Exam pointers

- Compute an elasticity from two (p, Q) points and **name** it (elastic/inelastic/unit) — the –1/3 example.
- Use the point-elasticity trick $E_D = (dQ/dp) (p/Q)$ on a **linear demand** and show elasticity **changes along the curve** (more elastic at higher price).
- Give the **sign rules** for cross-price (substitutes/complements) and income (normal/inferior) elasticities, with an example each.
- Solve a **unit-tax market** ($p_b = p_s + t$) for p_b , p_s , Q , compute **revenue** and **deadweight loss** — the t=€3 example.
- State and **explain** the two big tax results: **legal ≠ economic incidence** (doesn't matter who's taxed) and **incidence depends on relative elasticities** (inelastic side pays more); use the sin-tax application.
- Explain **why elasticity is unit-free** and why demand is **more elastic in the long run** and for narrower product categories.

Bridge. Lecture 3 turns the Chapter 2 model from *qualitative* ("the curve shifts left") into *quantitative* ("by how much, and split how between price and quantity"). Elasticity is also the hinge to later chapters: **firm pricing** (a monopolist's markup depends on E_D), **tax policy & deadweight loss** (Ch. 7), and **who gains from trade**.

★ Fun facts & memorable details (Lecture 3)

Sticky bits from Elasticities & Taxes.

- **It doesn't matter who you tax.** In a competitive market the burden of a tax is **identical** whether the government bills the shop or the shopper — "all that matters is that the tax drives a wedge." (This is why arguing about *who* a tax is "on" often misses the point.)
- **Why cigarette & alcohol taxes raise so much money:** demand is **inelastic**, so people keep buying — the tax is **passed to consumers** with little drop in quantity (and is often **regressive**).
- **The soda-tax number:** the own-price elasticity of sugary drinks (≈ -1.37) is a real input to real policy — a 5% price rise cuts consumption $\sim 7\%$.
- **Engel's law (1857):** the **richer** you get, the **smaller** the share of income you spend on **food** — one of the oldest empirical regularities in economics, and still used to gauge living standards.
- **Pharma = "defensive stock":** demand for medicine barely moves with income (income-inelastic), so drug companies hold up in recessions.
- **Elasticity has a time dimension:** when petrol jumps, you can't do much *this week* (inelastic), but over years you buy a smaller car and move closer to work (elastic) — the "short-run vs long-run" split.
- **Brand vs category:** demand for *a* beer is wildly elastic (≈ -4.7 — switch brands!), but demand for *beer* is inelastic (≈ -0.6) — same drink, opposite elasticity, just different zoom level.
- **Swedes really do drive to Denmark for cheap booze** — enough that economists measured the cross-border **cross-price elasticity** of spirits ($+0.3$) from the data (Asplund et al. 2007).